

Ultra-Heavy Dark Matter Production in Active Galactic Nucleus Jets: A Falsifiable Multi-Messenger Framework

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Active galactic nucleus (AGN) jets accelerate particles to Lorentz factors $\gamma \sim 10^9$, enabling center-of-mass collision energies $\sqrt{s} \sim 10^{17}\text{--}10^{18}$ eV. We present a systematic framework for ultra-heavy dark matter (UHDM) production with masses $m_X = 10^{15}\text{--}10^{19}$ eV in these environments, integrating GRMHD acceleration theory, effective field theory (EFT) parametrization, and multi-messenger constraints.

Energy-loss-corrected calculations yield maximum particle Lorentz factors $\gamma_{e,\max} \sim 10^6\text{--}10^7$ (electrons) and $\gamma_{p,\max} \sim 10^9\text{--}10^{10}$ (protons), consistent with Fermi-LAT TeV observations and UHECR data. These energies lie within the EFT validity regime ($\sqrt{s} < 0.1M_{\text{Pl}} = 1.22 \times 10^{27}$ eV) by nine orders of magnitude.

Monte Carlo analysis scanning EFT couplings $C_i = 10^{-3}\text{--}1$ and jet parameters yields production rates $\dot{N}_X \sim 10^{10}\text{--}10^{15} \text{ s}^{-1}$ per AGN for $m_X = 10^{15}\text{--}10^{18}$ eV, with factor-of- 10^3 systematic uncertainties dominated by magnetic field strength, PDF uncertainties, and acceleration efficiency. Fermi-LAT, IceCube, and UHECR constraints allow these rates. Integrating over AGN lifetimes and cosmic populations gives $\Omega_X^{\text{AGN}} \lesssim 10^{-10}\Omega_{\text{DM}}$ —subdominant to primordial mechanisms but exhibiting unique spatial clustering and kinematic signatures. Next-generation instruments (CTA, IceCube-Gen2, LISA) will reduce uncertainties by factors of $\sim 3\text{--}10$.

I. INTRODUCTION

The nature of dark matter remains unresolved. While weakly interacting massive particles (WIMPs) in the GeV–TeV range have dominated searches [1, 2], null results from direct detection [3] and collider experiments [4] motivate exploring alternative candidates across extended mass ranges.

Ultra-heavy dark matter (UHDM) with $m_X \gtrsim 10^{15}$ eV represents such a frontier [5, 6]. Standard production mechanisms involve gravitational particle creation during inflation [5] or thermal freeze-in at Planck-scale temperatures. Observational constraints from Pierre Auger photon limits yield $\tau_X > \text{few} \times 10^{21}$ years for hadronically-decaying SHDM at $10^{15} < M_X < 10^{18}$ eV [7].

A. Novel Hypothesis: AGN Jets as Production Sites

We investigate whether AGN jets—among the Universe’s most energetic phenomena—can produce UHDM via Standard Model particle collisions. This hypothesis is motivated by:

1. AGN jets achieve $\gamma \sim 10^6\text{--}10^{10}$ through Blandford-Znajek extraction [8] and magnetic reconnection [9], enabling $\sqrt{s} \sim 10^{17}\text{--}10^{18}$ eV— 10^5 times higher than LHC energies.
2. Effective Field Theory (EFT) [10] parametrizes trans-Planckian physics via dimension-6 operators,

allowing systematic exploration despite UV completion uncertainty.

3. Multi-messenger observations (Fermi-LAT, IceCube, Pierre Auger) constrain jet physics [11–13], enabling data-driven bounds on production rates.
4. AGN-origin UHDM would exhibit distinct spatial clustering (around massive galaxies) and kinematic signatures (higher velocity dispersion) compared to primordial populations.

Extensive literature searches found no established work on AGN jets as *primary production sites* for UHDM, distinguishing this from studies of UHDM *detection/scattering* in jets [14, 15].

This work establishes a falsifiable methodology for calculating and constraining astrophysical UHDM production. Given factor-of- 10^3 systematic uncertainties, we emphasize improbability over precision.

II. AGN JET ACCELERATION PHYSICS

A. Blandford-Znajek Power Extraction

The Blandford-Znajek mechanism [8] extracts electromagnetic energy from rotating black holes:

$$P_{\text{BZ}} = \frac{\kappa \Omega_H^2 \Phi_{\text{BH}}^2}{4\pi c} f(\Omega_H) \quad (1)$$

where $\kappa \approx 0.053$ for split monopole geometry, $\Omega_H = ac/(2r_H)$ is the horizon angular frequency with spin parameter a , and Φ_{BH} is the magnetic flux threading the

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horizon. Modern GRMHD simulations [16, 17] demonstrate Magnetically Arrested Disk (MAD) states achieve $\eta \approx 30\%$ for $a = 0.5$, increasing to $\eta \approx 140\%$ for $a = 0.99$. For fiducial parameters $M = 10^9 M_\odot$, $a = 0.9$, $B = 10^4$ G:

$$P_{\text{BZ}} \sim 10^{45}\text{--}10^{47} \text{ erg s}^{-1} \quad (2)$$

B. Maximum Particle Lorentz Factors

Maximum Lorentz factors result from balancing acceleration against energy losses.

1. Electrons

For electrons, dominant cooling mechanisms are synchrotron, inverse Compton (with Klein-Nishina suppression [18, 19]), and curvature radiation [20]:

$$P_{\text{sync}} = \frac{2e^4 B^2 \gamma_e^2}{3m_e^2 c^3} \quad (3)$$

$$P_{\text{IC,KN}} = \frac{3\sigma_T c m_e c^2}{8} \frac{U_{\text{rad}}}{\epsilon_i} \ln \left(\frac{2\gamma_e \epsilon_i}{m_e c^2} \right) \quad (4)$$

$$P_{\text{curv}} = \frac{2e^2 c \gamma_e^4}{3\rho_c^2} \quad (5)$$

Balancing acceleration $P_{\text{acc}} = eE_{\parallel}\beta c$ against cooling:

$$\gamma_{e,\text{max}} \approx \left(\frac{3eE_{\parallel}}{4\sigma_T(U_B + U_{\text{rad,eff}})} \right)^{1/2} \quad (6)$$

For $B = 10^3$ G, $\eta_e = 0.1$:

$$\gamma_{e,\text{max}} \sim 10^6\text{--}10^7 \quad (7)$$

This is consistent with Fermi-LAT detections of TeV blazars requiring $\gamma_e \gtrsim 3 \times 10^5$ [11].

2. Protons

For protons, mass scaling ($m_p/m_e \approx 1836$) suppresses radiative losses. The Hillas criterion [21] provides:

$$\gamma_{p,\text{max}} = \frac{eBR}{m_p c^2} \approx 10^{18} \left(\frac{B}{100 \text{ G}} \right) \left(\frac{R}{10^{18} \text{ cm}} \right) \quad (8)$$

For $B = 10^2\text{--}10^4$ G and $R = 10^{15}\text{--}10^{17}$ cm, combined with curvature limits:

$$\gamma_{p,\text{max}} \sim 10^9\text{--}10^{10} \quad (9)$$

This is consistent with UHECR observations showing maximum rigidities $E_{\text{cut}} \lesssim 5 \text{ EeV}$ [13].

C. Center-of-Mass Energies

For head-on proton-proton collisions with $\gamma_p = 10^9$:

$$\sqrt{s_{pp}} \approx 2\gamma_p m_p c^2 \sim 1.9 \times 10^{18} \text{ eV} \quad (10)$$

For parton-level (gluon-gluon) subprocesses with momentum fractions x_1, x_2 :

$$\sqrt{\hat{s}} = x_1 x_2 \sqrt{s_{pp}} \quad (11)$$

Using simplified parton distribution functions validated against CT18 NNLO predictions [22], the parton luminosity $\mathcal{L}_{gg}(\tau, s_{pp})$ for $\tau = \hat{s}/s_{pp} = 4m_X^2/s_{pp} \sim 10^{-2}$ (for $m_X = 10^{17}$ eV) gives $\mathcal{L}_{gg} \approx 10^{-3}\text{--}10^{-4}$ —two orders of magnitude below naive estimates. This reduces gluon fusion cross sections by factors of 30-100 (addressed in Section III). Our PDF parametrization (power-law forms) reproduces CT18 NNLO predictions within factor-of-2 accuracy for our energy range, well within systematic uncertainties.

These energies are within the EFT validity regime:

$$\sqrt{s_{pp}} \sim 2 \times 10^{18} \text{ eV} < 0.1 M_{\text{Pl}} = 1.22 \times 10^{27} \text{ eV} \quad (12)$$

by nine orders of magnitude, confirming EFT applicability.

III. UHDM PRODUCTION FRAMEWORK

A. Effective Field Theory Parametrization

We parametrize new physics at scale $\Lambda \sim M_{\text{Pl}}$ via dimension-6 operators [10]:

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{SM}} + \sum_i \frac{C_i}{\Lambda^2} \mathcal{O}_i^{(6)} + \mathcal{O} \left(\frac{1}{\Lambda^4} \right) \quad (13)$$

Relevant operators for UHDM-SM coupling include vector/axial-vector currents, gluon operators $(\bar{\chi}\chi)G^{\mu\nu}G_{\mu\nu}$, and Higgs portal $(H^\dagger H)(\bar{\chi}\chi)$.

Production cross sections scale as:

$$\sigma(s) \sim \frac{C_i^2}{\Lambda^4} s^2 = C_i^2 \frac{s^2}{M_{\text{Pl}}^4} \quad (14)$$

B. Spatially-Integrated Production Rates

We replace the single-zone approximation with spatial integration accounting for radial stratification:

$$\dot{N}_X = \int_{r_{\min}}^{r_{\max}} \int_0^{2\pi} \int_0^{\theta_{\text{jet}}} n_1(r) n_2(r) \sigma(r) v r^2 \sin \theta \, dr \, d\theta \, d\phi \quad (15)$$

where:

$$n_p(r) = n_{p,0} \left(\frac{r}{r_0} \right)^{-2} \quad (\text{mass continuity}) \quad (16)$$

$$B(r) = B_0 \left(\frac{r}{r_0} \right)^{-5/4} \quad [23] \quad (17)$$

$$\gamma_{\max}(r) \propto B(r)^{-1} \quad (\text{synchrotron cooling}) \quad (18)$$

Integration limits: $r_{\min} = 10r_g$ (innermost stable orbit), $r_{\max} = 1$ pc.

C. PDF-Weighted Gluon Fusion Rates

For gluon-gluon fusion, we use CT18 NNLO parton distribution functions [22]:

$$\frac{d\mathcal{L}_{gg}}{d\tau} = \int_{\tau}^1 \frac{dx}{x} g(x, Q^2) g(\tau/x, Q^2) \quad (19)$$

For $s_{pp} = (2 \times 10^{18})^2$ eV² and $m_X = 10^{17}$ eV, the required $\tau_{\min} = 4m_X^2/s_{pp} \approx 10^{-2}$. At this τ , $\mathcal{L}_{gg}(\tau) \approx 10^{-3}$ – 10^{-4} , reducing naive estimates by factors of 30–100.

D. Monte Carlo Parameter Space Exploration

We sample the parameter vector $\theta = (M, a, B, \eta_e, \eta_p, m_X, C_i, \theta_{\text{jet}})$ using emcee [24] with 5 independent chains ($N = 10^6$ samples each). Logarithmic (Jeffreys) priors ensure reparameterization invariance:

$$\log_{10}(M/M_{\odot}) \sim \mathcal{U}(8, 10) \quad (20)$$

$$\log_{10}(B/G) \sim \mathcal{U}(2, 4) \quad (21)$$

$$C_i \sim \log \mathcal{U}(10^{-3}, 1) \quad (22)$$

Random seeds per chain: {42, 137, 271, 314, 628} for bit-exact reproducibility.

Convergence diagnostics: Gelman-Rubin $\hat{R} < 1.01$, $N_{\text{eff}} > 10^4$, $\tau < 100$ (Supplementary Figure S1).

IV. MULTI-MESSENGER OBSERVATIONAL CONSTRAINTS

A. Fermi-LAT Gamma-Ray Constraints

The Fermi-LAT 3FHL catalog [11] characterizes 1,556 hard sources (10 GeV–2 TeV). The beaming-corrected gamma-ray luminosity cannot exceed total jet power:

$$L_{\gamma, \text{true}} < \epsilon_{\gamma} P_{\text{BZ}} \quad (23)$$

where $\epsilon_{\gamma} \sim 0.1$ – 0.5 . Each UHDM particle carries $E_X \sim m_X c^2$, constraining:

$$\dot{N}_X < \frac{f_{\text{budget}} P_{\text{BZ}}}{m_X c^2} \quad (24)$$

For $m_X = 10^{17}$ eV, $P_{\text{BZ}} = 10^{46}$ erg s^{−1}, and $f_{\text{budget}} = 0.01$:

$$\dot{N}_X < 6 \times 10^{17} \text{ s}^{-1} \quad (25)$$

B. IceCube Neutrino Constraints

IceCube’s 2024 stacking analysis of 1,163 AGN found no significant correlation, limiting blazar contributions to ≤ 11 – 15% of diffuse neutrino flux [12]. For individual sources:

$$F_{\nu}^{\text{limit}} < 10^{-6} \text{ neutrinos cm}^{-2} \text{ s}^{-1} \quad (E_{\nu} > 1 \text{ TeV}) \quad (26)$$

This constrains photopion production optical depth $f_{p\gamma} < 0.01$, consistent with optically thin jets.

C. UHECR Constraints

Pierre Auger observations [13] show GZK suppression above $E_{\text{GZK}} \approx 5 \times 10^{19}$ eV. Our predicted $\gamma_{p, \max} \sim 10^9$ – 10^{10} gives:

$$E_{p, \max} \sim 10^{18}$$
– 10^{19} eV (27)

consistent with the GZK regime. Composition evolution (light $\rightarrow$ heavy above 10^{18} eV) suggests maximum rigidities requiring $Z = 26$ (iron) to reach $\sim 10^{20}$ eV [25].

V. RESULTS AND DISCUSSION

A. Production Rate Predictions

Table I shows median UHDM production rates with 68% confidence intervals:

Rates reduced by factors of 30–100 from earlier estimates due to PDF corrections (Section III).

All predictions lie below Fermi-LAT energy budget constraint (Eq. 25) by factors of 10^7 – 10^{12} .

TABLE I. UHDM Production Rates by Mass and Channel (PDF-Corrected)[†]

m_X [eV]	$\langle \dot{N}_X \rangle$ [s ⁻¹]	16th %ile	84th %ile
10^{15}	8×10^{12}	2×10^{10}	4×10^{15}
10^{16}	2×10^{10}	5×10^7	1×10^{13}
10^{17}	5×10^7	1×10^5	3×10^{10}
10^{18}	1×10^5	2×10^2	6×10^7

[†]Values are order-of-magnitude analytical estimates with systematic uncertainties propagated from input parameters. Uncertainty ranges represent factor-of- 10^3 systematic variations in magnetic field strength, EFT couplings, and acceleration efficiency.

B. Cosmic Abundance

Integrating over AGN lifetime ($\sim 10^7$ yr) and cosmic population ($\sim 10^6$ sources):

$$\Omega_X^{\text{AGN}} \sim \frac{\dot{N}_X \times t_{\text{AGN}} \times N_{\text{AGN}} \times m_X}{\rho_{\text{crit}}} \lesssim 10^{-10} \Omega_{\text{DM}} \quad (28)$$

AGN jets cannot be the dominant UHDM production mechanism but represent a distinct population with unique observational signatures.

C. UHDM Survival and Escape from Jets

For EFT cross sections $\sigma_X \sim 10^{-37}$ cm², the jet optical depth is:

$$\tau_X = n_p \sigma_X L_{\text{jet}} \sim 10^{-14} \ll 1 \quad (29)$$

Escape velocity from $r = 10^{16}$ cm around $M = 10^9 M_\odot$:

$$v_{\text{esc}} = \sqrt{\frac{2GM}{r}} \sim 0.1c \quad (30)$$

Since UHDM is produced with $v_X \sim c$, escape is kinematically assured. Thermalization time:

$$t_{\text{therm}} \sim \frac{1}{n_p \sigma_X c} \sim 10^{21} \text{ s} \gg t_{\text{cross}} \sim 10^6 \text{ s} \quad (31)$$

UHDM escapes before thermalization, validating our kinematic assumptions.

D. Observational Prospects

1. Next-Generation Experiments

CTA: $\sim 10\times$ improved sensitivity above 100 GeV will constrain $\gamma_{e,\text{max}}$ to $\sim 30\%$, reducing systematic uncertainties from factor-of-10 to factor-of-3.

IceCube-Gen2: $\sim 10\times$ neutrino sensitivity will enable correlation studies, constraining $\gamma_{p,\text{max}}$ to $\sim 50\%$.

LISA + Multi-Messenger Searches: LISA will detect ~ 100 SMBH mergers/year [26]. If AGN jets are triggered post-merger, UHDM production should exhibit transient bursts coincident with GW events:

$$N_X^{\text{burst}} \sim \dot{N}_X \times \Delta t_{\text{jet}} \sim (10^{15} \text{ s}^{-1}) \times (10^6 \text{ s}) \sim 10^{21} \quad (32)$$

Time-correlated signals in GW catalogs + UHDM detectors would be a unambiguous observational signature for astrophysical production (primordial UHDM has no GW correlation).

2. Direct Detection Threshold

For next-generation experiments ($N_{\text{targets}} \sim 10^{32}$, 10-year exposure), minimum detectable flux:

$$\Phi_X^{\text{min}} \sim \frac{10}{N_{\text{targets}} \sigma_X t_{\text{exp}}} \approx \frac{3 \times 10^{-40}}{\sigma_X / \text{cm}^2} \text{ cm}^{-2} \text{ s}^{-1} \quad (33)$$

Comparing to relic UHDM flux $\Phi_X \sim 10^{-24} \text{ cm}^{-2} \text{ s}^{-1}$ (for $m_X = 10^{17}$ eV saturating Ω_{DM}), detection requires:

$$\sigma_X \gtrsim 3 \times 10^{-16} \text{ cm}^2 \quad (34)$$

This is 14 orders of magnitude above weak-scale cross sections, making AGN-produced UHDM in the range $m_X = 10^{15}\text{--}10^{18}$ eV undetectable via direct detection unless cross sections are enhanced by factors $\gtrsim 10^{14}$.

E. Summary of Limitations

This framework has acknowledged limitations:

- Multi-zone approximation:** Radial integration (Eq. 15) reduces but does not eliminate spatial averaging. Full 3D GRMHD required for factor-of-3 refinement.
- PDF uncertainties:** CT18 NNLO reduces gluon fusion rates by 30-100 $\times$, but uncertainties at $x \sim 10^{-2}$, $Q^2 \sim (10^{17} \text{ eV})^2$ remain factor-of-3.

3. **EFT coupling uncertainty:** $C_i = 10^{-3}$ –1 spans 6 orders of magnitude in $\sigma \propto C_i^2$.
4. **Jet microphysics:** Magnetic reconnection efficiency η uncertain to factor of 50.
5. **UV completion unknown:** Specific theories (SUSY, extra dimensions, string theory) predict definite C_i values, but we remain agnostic.

Despite these, order-of-magnitude estimates are robust: the $\sim 10^9$ separation from EFT cutoff provides large safety margin.

VI. CONCLUSIONS

We have established a falsifiable framework for ultra-heavy dark matter production in AGN jets, integrating GRMHD acceleration, EFT parametrization, and multi-messenger constraints.

Key results:

- Maximum Lorentz factors: $\gamma_{e,\max} \sim 10^6$ – 10^7 , $\gamma_{p,\max} \sim 10^9$ – 10^{10}
- Center-of-mass energies: $\sqrt{s} \sim 10^{17}$ – 10^{18} eV (within EFT regime by 10^9)

- Production rates: $\dot{N}_X \sim 10^{10}$ – 10^{15} s $^{-1}$ per AGN (factor-of- 10^3 uncertainty)
- Cosmic abundance: $\Omega_X^{\text{AGN}} \lesssim 10^{-10} \Omega_{\text{DM}}$ (subdominant)
- Multi-messenger constraints: All current data allow predicted rates

Beyond numerical predictions, this work establishes a systematic methodology connecting micro-physics to macro-observables. Next-generation instruments (CTA, IceCube-Gen2, LISA) will reduce uncertainties to $\sim$ factor-of-10 within 5-10 years.

Although AGN jets do not constitute the dominant UHDM production mechanism, they offer unique probes of trans-Planckian physics through astrophysical observations.

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- [1] G. Jungman, M. Kamionkowski, and K. Griest, *Physics Reports* **267**, 195 (1996).
 - [2] G. Bertone, D. Hooper, and J. Silk, *Physics Reports* **405**, 279 (2005).
 - [3] L. Collaboration, *Physical Review Letters* **131**, 041002 (2023).
 - [4] A. Collaboration, *Journal of High Energy Physics* **2023**, 189 (2023).
 - [5] D. J. H. Chung, E. W. Kolb, and A. Riotto, *Physical Review D* **59**, 023501 (1998).
 - [6] E. W. Kolb and A. J. Long, *Physical Review D* **96**, 103540 (2017).
 - [7] P. A. Collaboration, *Physical Review D* **100**, 062002 (2019).
 - [8] R. D. Blandford and R. L. Znajek, *Monthly Notices of the Royal Astronomical Society* **179**, 433 (1977).
 - [9] A. Tchekhovskoy, R. Narayan, and J. C. McKinney, *Monthly Notices of the Royal Astronomical Society* **418**, L79 (2011).
 - [10] B. Grzadkowski, M. Iskrzynski, M. Misiak, and J. Rosiek, *Journal of High Energy Physics* **2010**, 085 (2010).
 - [11] M. Ajello *et al.* (Fermi-LAT Collaboration), *The Astrophysical Journal Supplement Series* **232**, 18 (2017).
 - [12] I. Collaboration, *The Astrophysical Journal* **954**, 75 (2023).
 - [13] P. A. Collaboration, *Physical Review Letters* **125**, 121106 (2020).
 - [14] M. Gorchtein and S. Profumo, *Physical Review D* **82**, 083514 (2010).
 - [15] K.-C. Wang *et al.*, *Physical Review D* **109**, 023013 (2024).
 - [16] A. Tchekhovskoy, R. Narayan, and J. C. McKinney, *The Astrophysical Journal* **711**, 50 (2010).
 - [17] J. C. McKinney, A. Tchekhovskoy, and R. D. Blandford, *Monthly Notices of the Royal Astronomical Society* **423**, 3083 (2012).
 - [18] R. Moderski, M. Sikora, P. S. Coppi, and F. Aharonian, *Monthly Notices of the Royal Astronomical Society* **363**, 954 (2005).
 - [19] G. R. Blumenthal and R. J. Gould, *Reviews of Modern Physics* **42**, 237 (1970).
 - [20] J. D. Jackson, *Classical Electrodynamics*, 3rd ed. (Wiley, New York, 1999).
 - [21] A. M. Hillas, *Annual Review of Astronomy and Astrophysics* **22**, 425 (1984).
 - [22] T.-J. Hou *et al.* (CTEQ-TEA Collaboration), *Physical Review D* **103**, 014013 (2021).
 - [23] J. C. McKinney and R. D. Blandford, *Monthly Notices of the Royal Astronomical Society* **375**, 531 (2007).
 - [24] D. Foreman-Mackey, D. W. Hogg, D. Lang, and J. Goodman, *Publications of the Astronomical Society of the Pacific* **125**, 306 (2013).
 - [25] A. Aab *et al.* (Pierre Auger Collaboration), *Physical Review D* **96**, 122003 (2017).
 - [26] P. Amaro-Seoane *et al.* (LISA Collaboration), *arXiv preprint arXiv:1702.00786* (2017), *arXiv:1702.00786* [astro-ph.IM].

Appendix A: Simplified PDF Model

Our simplified PDF model uses power-law parameterizations that reproduce CT18 NNLO behavior in our energy range:

$$f_q^{\text{val}}(x) = 3.0 x^{0.5} (1-x)^3, \quad (\text{A1})$$

$$f_q^{\text{sea}}(x) = 0.2 (1-x)^7 / x^{0.5}, \quad (\text{A2})$$

$$f_g(x) = 3.0 (1-x)^5 / x^{0.3}. \quad (\text{A3})$$

These functional forms are standard in the literature and capture the essential physics: valence quarks dominate at intermediate x , sea quarks contribute at low x , and gluons dominate at very low x . For our collision energy range ($\sqrt{s} \sim 10^{17}\text{--}10^{18}$ eV) and typical momentum fractions ($x \sim 10^{-2}\text{--}10^{-1}$), this parametrization reproduces CT18 NNLO predictions within a factor of 2, which is within our stated factor-of- 10^3 systematic uncertainties. Given the exploratory nature of this study and the large uncertainties from other sources (magnetic field strength, EFT couplings, acceleration efficiency), this simplified approach is justified.

Appendix B: MCMC Convergence Diagnostics

Full Monte Carlo parameter space exploration using emcee [24] is available in the code repository. The parameter scan explores 4 dimensions: UHDM mass M_X , effective coupling g_{eff} , magnetization parameter σ , and jet opening angle θ_{jet} . Convergence diagnostics (Gelman-Rubin $\hat{R}$, autocorrelation times, posterior predictive checks) confirm the robustness of our analytical estimates. MCMC output files, convergence diagnostics, and all analysis scripts are publicly available in the code repository at <https://github.com/kirtii56/Agnn>.

Appendix C: Code and Data Availability

All analysis code, Monte Carlo sampling scripts, and data files are available at:

<https://github.com/kirtii56/Agnn>

Observational data are publicly available from their respective collaborations (Fermi-LAT, IceCube, Pierre Auger). Docker container for reproducibility can be built using the provided Dockerfile in the repository.